United International University

Department of Computer Science and Engineering

Software Engineering Assignment

Course Code: CSE 3421 – Software Engineering

Total Marks: 20

---

1. Critical Path Method (10 Marks)

For the following project tasks, calculate the **Earliest Start (ES), Earliest Finish (EF), Latest Start (LS), and Latest Finish (LF)** for every task. Determine the project's earliest completion time and identify the **critical path** using the Critical Path Method (CPM).

Task	Duration	Predecessor(s)
A	10	–
B	5	A
C	8	B
D	11	A, C
E	15	–
F	5	D
G	7	D, E
H	12	–
I	9	G, H
J	6	I
K	11	C, I

2. Smart Home Device Management System – Design Patterns (10 Marks)

A technology company is developing a Smart Home Management System that lets users control devices from Samsung, Xiaomi, and LG. Each manufacturer provides a family of compatible smart lights, thermostats, and security cameras. Although these products perform similar functions, each manufacturer's devices have different implementations and communication protocols.

When a user selects a manufacturer as the home ecosystem, the application must create the corresponding devices from that manufacturer; devices from different manufacturers must not be mixed. The application should use general interfaces for lights, thermostats, and cameras, rather than depend directly on manufacturer-specific classes. The company plans to support additional manufacturers in the future.

a) Identify the most suitable design pattern and briefly explain why it fits the requirements. (2 Marks)

b) Draw a UML class diagram for your design. Show the client, factory interface, concrete manufacturer factories, product interfaces, and manufacturer-specific products. Include relevant creation methods and clearly show the relationships. (6 Marks)

c) Briefly explain how the design allows a new manufacturer to be added without major changes to the existing application logic. (2 Marks)